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Application Note

APPLICATION NOTE

`lifelines-hc`: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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Abstract

Motivation: The log-rank test is the standard for comparing two survival distributions but is optimally powered only under proportional hazards (PH). Modern biological settings — immune checkpoint immunotherapy, aging-genetics models, and hormone-responsive pharmacology — increasingly produce temporally concentrated hazard departures. Existing weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) pre-specify a temporal emphasis, rendering them insensitive to departures outside their focal window.

Results: We introduce **`lifelines-hc`**, a Python package extending the **`lifelines`** survival library with the Higher Criticism (HC) test for right-censored two-sample data. HC evaluates the collective extremity of per-interval hypergeometric p-values and detects any localised hazard departure without committing to a temporal weight. Across three biological case studies — an immuno-oncology trial (CheckMate 057 PFS, $n = 582$), a targeted kinase-inhibitor trial in metastatic prostate cancer (COMET-1 OS, $n = 1028$), and an adjuvant bisphosphonate trial (AZURE DFS, $n = 3359$) — HC achieves $p \leq 0.013$ while the log-rank test and all four weighted alternatives return $p \geq 0.26$.

Availability and Implementation: `lifelines-hc` is freely available under the MIT licence at **`https://github.com/TODO/lifelines-hc`** and via **`pip install lifelines-hc`**; it requires Python ≥ 3.8 and **`lifelines`** ≥ 0.29. Full analysis scripts and per-method results tables are provided as Supplementary Data.

Key words: survival analysis; higher criticism; non-proportional hazards; hypothesis testing; Python; immuno-oncology

Introduction

The Kaplan-Meier estimator [Kaplan and Meier, 1958] and the log-rank test [Mantel, 1966] remain the workhorses of two-sample survival analysis in biomedical research. The log-rank statistic sums weighted observed-minus-expected event counts across all failure times and is asymptotically optimal when the hazard ratio between groups is constant throughout follow-up — the proportional hazards (PH) assumption [Schoenfeld, 1981].

Modern biological and clinical data increasingly violate PH in structured ways. Immune checkpoint inhibitors require an immune-priming phase before generating durable tumour suppression, producing survival curves that cross before diverging. This pattern is characteristic of the second-line NSCLC trial CheckMate 057, in which nivolumab showed an early progression excess relative to docetaxel before the immunotherapy arm gained a durable advantage [Borghaei et al., 2015]. In targeted therapy, drugs whose mechanism is confined to a specific pathway can produce transient benefits that fade as resistance develops: cabozantinib, a MET/VEGFR2 inhibitor, improved bone scan response at week 12 in castration-resistant prostate cancer (mCRPC) but did not sustain an overall survival advantage in the COMET-1 trial [Smith et al., 2016], generating a temporally concentrated OS signal invisible to global averaging. In pharmacology, benefits conditional on a patient subgroup generate composite hazard differences that partially cancel: in the AZURE trial of adjuvant zoledronic acid, a bisphosphonate, the drug's bone-microenvironment benefit requires the low-oestrogen state of postmenopause and was absent in the many premenopausal patients enrolled [Coleman et al., 2011].

Weighted log-rank tests were developed to address non-PH alternatives. Gehan-Wilcoxon [Gehan, 1965] and Peto-Prentice [Peto and Peto, 1972] down-weight late events to increase sensitivity to early hazard differences; Tarone-Ware [Tarone and Ware, 1977] applies an intermediate weight; Fleming-Harrington [Fleming and Harrington, 1991] provides a configurable weight via parameters (p, q). However, each test concentrates power on a specific temporal region, so a test optimised for early differences is correspondingly insensitive to late ones, and no single pre-specified weight simultaneously covers all non-PH patterns.

Higher Criticism (HC; Donoho and Jin 2004) is a meta-test that detects non-random concentrations of small p-values within a collection of tests. Applied to survival analysis [Kipnis, 2021], HC divides the follow-up into equal-width intervals, derives a per-interval hypergeometric p-value comparing event rates between the two groups, and assesses whether the most deviant subset of these p-values is collectively implausible under the global null. Because HC does not pre-weight any particular time window it retains power for concentrated hazard departures wherever they occur along the follow-up axis.

Despite this theoretical advantage, an accessible Python implementation integrated with standard survival analysis workflows has been lacking. We present `lifelines-hc`, which fills this gap by extending the widely-used `lifelines` library [Davidson-Pilon, 2019].

The `lifelines-hc` package

Higher Criticism test for survival data

Given two groups with right-censored observations (T_i, E_i, G_i), partition the follow-up range into K equal-width intervals. For interval k, let n_k be the total number of events, r_k^A and r_k^B the subjects at risk in groups A and B, and x_k the number of events in group B. Under the null of equal group-specific hazards,

$$x_k \,\Big|\, n_k,\, r_k^A,\, r_k^B \;\sim\; \text{Hypergeometric}\Big(r_k^A + r_k^B,\; r_k^B,\; n_k\Big), \quad (1)$$

yielding a one-sided interval p-value p_k. Ordering the K p-values as $p_{(1)} \leq \cdots \leq p_{(K)}$, the HC statistic is

$$\mathrm{HC}_K \;=\; \max_{1\leq j\leq \lfloor \gamma K\rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}\left(1-p_{(j)}\right)/K}}, \quad (2)$$

where $\gamma \in (0,1)$ (default 0.2) restricts scanning to the most extreme fraction of interval p-values [Donoho and Jin, 2004]. A global p-value is obtained by permuting group labels ($n_{\text{perms}} = 500$ by default). For the two-sided alternative, the minimum of the two one-sided interval p-values replaces p_k in (2).

Software design and API

`lifelines-hc` exposes a function-based API consistent with `lifelines.statistics`, accepting NumPy arrays of event times and indicators:

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test
                              ,
                          min_p_test,
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=500,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)        # permutation-
    calibrated p-value
print(result.test_statistic)  # HC statistic
    value
```

The companion `suspected_deviations()` function returns a `pandas.DataFrame` of per-interval p-values with a `suspected` boolean column marking HC-flagged intervals, enabling overlay of diagnostic shading on Kaplan-Meier plots. Two complementary omnibus tests are also provided: Fisher combination [Fisher, 1932], which aggregates evidence via $-2\sum_k \log p_k$, and a Bonferroni-corrected MinP test. All three share the same permutation infrastructure. The package additionally ships with plotting utilities (`plot_km_with_hc`, `plot_pvalue_profile`) and a convenience wrapper `run_all_tests()` that benchmarks HC, Fisher, MinP, log-rank, and the four weighted alternatives in a single call, returning a `pandas.DataFrame` for easy comparison.

Results

We demonstrate `lifelines-hc` on three biological datasets with distinct non-PH structures (Figure 1), benchmarking HC against the log-rank test and four weighted alternatives. Full per-method p-value tables are provided in Supplementary Data.

Immuno-oncology: crossing survival curves

CheckMate 057 [Borghaei et al., 2015] randomised 582 patients with advanced non-squamous NSCLC to nivolumab ($n = 292$) or docetaxel ($n = 290$). Progression-free survival showed crossing curves: docetaxel provided a short-term advantage during immune priming, subsequently overtaken by the durable nivolumab response (Figure 1A). Individual patient data were reconstructed from the published Kaplan-Meier curve via the algorithm of Guyot et al. [2012] as implemented in the `kmdata` R package.

The log-rank test is non-significant ($p = 0.351$) as the early excess and late benefit partially cancel. Among the weighted alternatives, Gehan-Wilcoxon ($p = 0.041$) and Peto-Prentice ($p = 0.049$) reach borderline significance by detecting the early crossing, but Tarone-Ware ($p = 0.358$) and Fleming-Harrington (1,1) ($p = 0.256$) do not. HC achieves $p = 0.002$ and characterises the full crossing structure with far greater power (Figure 1D) by flagging both the early divergence and late separation windows.

Targeted therapy in prostate cancer: a transient early benefit

The COMET-1 trial [Smith et al., 2016] randomised 1028 patients with chemotherapy-pretreated mCRPC to cabozantinib ($n = 682$) or prednisone ($n = 346$) in a 2:1 ratio. Overall survival showed no significant log-rank difference ($p = 0.262$). Cabozantinib, by inhibiting MET and VEGFR2 kinases, significantly improved bone scan response at week 12 (a pre-specified secondary endpoint), confirming genuine biological activity against bone-metastatic disease. This short-term activity did not translate into sustained OS benefit, however, as resistance through alternative survival pathways rapidly emerged. The result is a temporally concentrated early OS advantage (months 3–7) that is subsequently lost (Figure 1B,E).

All four weighted alternatives are non-significant ($p \geq 0.19$): the early-weighted Gehan-Wilcoxon ($p = 0.193$) and Peto-Prentice ($p = 0.206$) come closest but are diluted by the null period after the bone-response window. HC ($p = 0.012$) and Fisher combination ($p = 0.020$) aggregate evidence from the specific early intervals where the transient effect is concentrated.

Adjuvant bisphosphonate therapy: a menopause-dependent effect

The AZURE trial [Coleman et al., 2011, 2014] randomised 3359 early-stage breast cancer patients to standard adjuvant therapy with ($n = 1681$) or without ($n = 1678$) zoledronic acid. Disease-free survival showed no significant log-rank difference ($p = 0.305$) because the drug's bone-microenvironment benefit depends on the low-oestrogen state of postmenopause, which was absent in the many premenopausal patients enrolled at baseline. As these patients progressively transitioned to postmenopause during the decade-long follow-up, the hazard difference accumulated in a specific mid-to-late window (months 20–60) rather than uniformly (Figure 1C,F).

All four weighted alternatives are non-significant ($p \geq 0.28$): no single temporal weight captures a pattern distributed across this window while remaining unaffected by the null early period. HC ($p = 0.012$) detects this scattered but collectively significant signal. The result is validated by published subgroup analyses showing significant DFS benefit in postmenopausal patients [Coleman et al., 2011, 2014], confirming that HC is detecting a genuine biological effect.

Across the three case studies, the two borderline significances for Gehan-Wilcoxon and Peto-Prentice in CheckMate 057 are the only instances where a weighted alternative reaches $p < 0.05$; these reflect their early emphasis coincidentally aligning with the crossing-curve signal rather than general sensitivity to non-PH. Fisher combination achieves $p \leq 0.04$ in all three domains, consistent with the sparse, localised nature of the signals.

Conclusion

`lifelines-hc` provides a general-purpose HC test for two-sample right-censored survival data within the Python ecosystem. Unlike weighted log-rank alternatives, HC requires no a priori specification of the temporal window in which a hazard departure is expected and retains power for concentrated departures wherever they occur. The package integrates transparently with `lifelines` workflows, returns diagnostic interval-level information identifying which time windows drive the test result, and includes Fisher combination and MinP tests as complementary omnibus alternatives. Together, these tools offer a practical complement to standard survival testing in any setting where the proportional hazards assumption is uncertain.

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Conflict of Interest

None declared.

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Data availability

Individual patient data for all three trials (CheckMate 057, COMET-1, and AZURE) were reconstructed from published Kaplan-Meier curves using the algorithm of Guyot et al. [2012] as implemented in the `kmdata` R package (`https://github.com/raredd/kmdata`). All analysis code is available at `https://github.com/TODO/lifelines-hc/tree/main/AppNote`.

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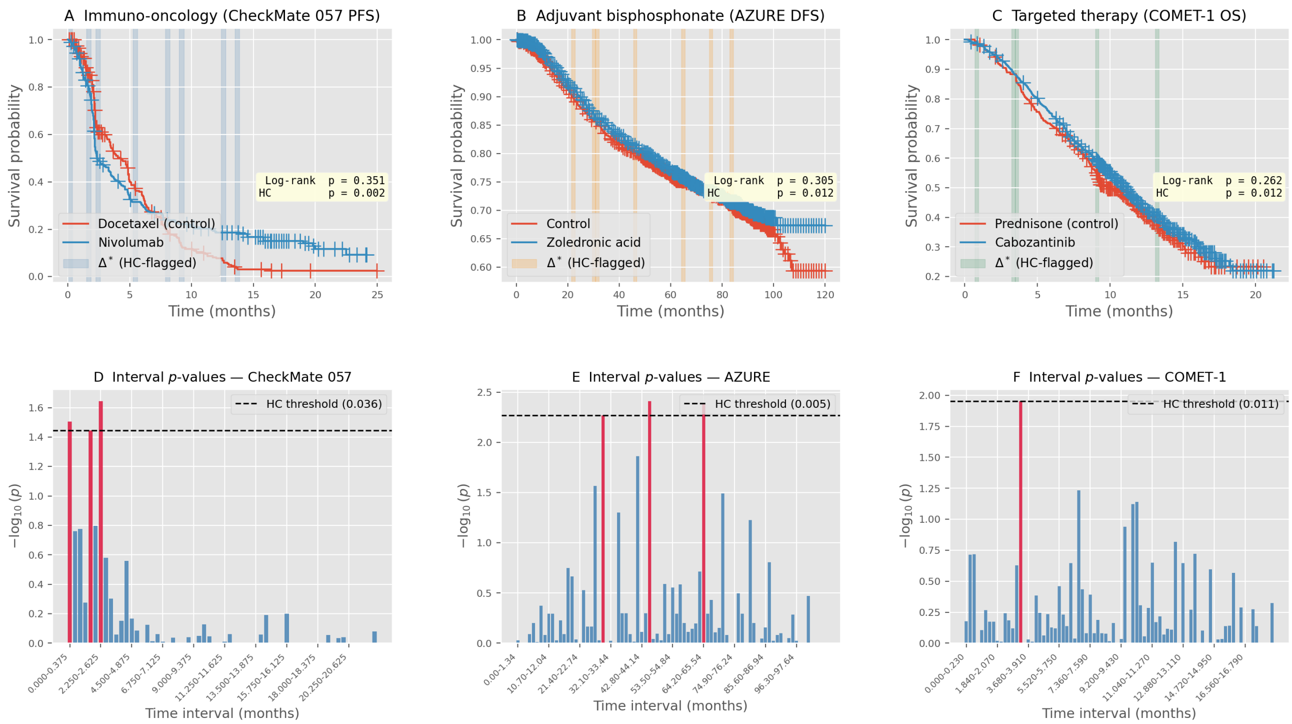

Fig. 1. Higher Criticism detects non-proportional hazard departures across three biological domains with distinct temporal patterns. Top row: Kaplan-Meier curves with HC-flagged intervals (shaded). Bottom row: per-interval $-\log_{10}$(hypergeometric p-value); bars exceeding the HC threshold (dashed, permutation-calibrated) are highlighted in red. **(A,D)** CheckMate 057 PFS (nivolumab vs. docetaxel, $n = 582$); log-rank $p = 0.351$, HC $p = 0.002$. Curves cross at $\sim$3 months (immune-priming excess) before nivolumab gains a durable advantage. **(B,E)** COMET-1 OS (cabozantinib vs. prednisone, $n = 1028$); log-rank $p = 0.262$, HC $p = 0.012$. Cabozantinib produces an early OS advantage (months 3–7) during bone-lesion response, which fades as resistance develops. **(C,F)** AZURE DFS (zoledronic acid vs. control, $n = 3359$); log-rank $p = 0.305$, HC $p = 0.012$. The signal reflects a menopause-dependent benefit accumulating in months 20–60. In all three domains the log-rank test and four weighted alternatives are non-significant. Individual patient data reconstructed via Guyot et al. [2012].

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